

Data Science Exam Solutions

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User prompt: solve this question as the following marks given exam ready

Response: Question 1: Mention a few sources of data impurities. Why is missing data not good? [5 Marks]

Sources of Data Impurities (2.5 Marks)

Data impurities (dirty data) arise at various stages of the data lifecycle. Key sources include:

- **Human Errors:** Manual data entry typos, misspellings, or selecting incorrect drop-down options during data collection.
- **Measurement & Instrument Errors:** Hardware malfunctions, faulty sensors, or calibration drifts yielding inaccurate readings.
- **Transmission & Network Glitches:** Network timeouts, packet loss, or system crashes during data integration or transfer.
- **System Integration Inconsistencies:** Combining disparate databases with conflicting schemas, naming conventions, or duplicate entity records.
- **Software Bugs:** Programming bugs in data transformation scripts or ETL (Extract, Transform, Load) pipelines.

Why Missing Data Is Not Good (2.5 Marks)

Missing data undermines analysis and downstream machine learning models in several ways:

- **Reduces Statistical Power:** Dropping rows with missing attributes decreases the overall sample size, which reduces the reliability of statistical conclusions.
- **Introduces Systemic Bias:** If values are not missing completely at random (e.g., higher-income individuals refusing to disclose earnings), dropping or naively imputing them skews model distributions.
- **Algorithm Incompatibility:** Most standard algorithms (e.g., standard neural networks, SVMs, linear regression) cannot process empty/null inputs directly.
- **Distorts Relationships:** Incomplete data impairs the ability to identify true correlations and patterns among attributes, often causing models to learn misleading signals.

Question 2: What are the different data attribute types? Write some examples. [5 Marks]

Data attributes (features/variables) describe specific characteristics of a data object and are broadly classified into two categories:

1. Qualitative (Categorical) Attributes (2.5 Marks)

These represent labels or distinct categories rather than quantities:

- **Nominal Attributes:** Values are categories or names with no inherent ordering or ranking.
 - *Examples:* Eye color (Blue, Brown, Green), Marital Status (Single, Married, Divorced), Blood Group (A , B , AB , O).
- **Binary Attributes:** A nominal attribute with exactly two states (0/1 or True/False).
 - *Symmetric:* Both outcomes are equally important (e.g., Gender: Male/Female).
 - *Asymmetric:* One state is rarer and more significant (e.g., Medical Test: Positive/Negative).
- **Ordinal Attributes:** Values have a meaningful rank or order, but the distance between consecutive values is not mathematically defined.
 - *Examples:* Customer Satisfaction rating (Poor, Fair, Good, Excellent), Education Level (High School, Bachelor's, Master's, PhD).

2. Quantitative (Numeric) Attributes (2.5 Marks)

These represent measurable quantities expressed as integers or real values:

- **Interval-Scaled Attributes:** Measured on an equal-interval scale, but lack a true/absolute zero point (ratios are not meaningful).
 - *Examples:* Temperature in Celsius or Fahrenheit (0°C does not indicate the absence of heat), Calendar dates.
- **Ratio-Scaled Attributes:** Numeric values with an absolute/meaningful zero point (a value of zero means absence of the attribute; ratios are meaningful).
 - *Examples:* Salary ($\$0$ means no income; $\$100\text{k}$ is twice $\$50\text{k}$), Age, Weight, Distance.

(Note: Numeric attributes can also be subdivided into **Discrete**—countable, e.g., customer count—and **Continuous**—uncountable real numbers, e.g., height).

Question 3: Quartiles are important in data discovery. How, and why? [5 Marks]

How Quartiles Work (2.5 Marks)

Quartiles divide a sorted numerical dataset into four equal parts using three cut-off points:

- **First Quartile (Q_1 / 25th percentile):** 25% of the data falls below this value.
- **Second Quartile (Q_2 / Median / 50th percentile):** Cuts the dataset directly in half.
- **Third Quartile (Q_3 / 75th percentile):** 75% of the data falls below this value.
- **Interquartile Range (IQR):** Calculated as $IQR = Q_3 - Q_1$, measuring the spread of the middle 50% of the data.

- **Boxplot Visualization (Five-Number Summary):** Quartiles form the foundation of the 5-number summary (*Minimum*, Q_1 , *Median*, Q_3 , *Maximum*). Outliers are explicitly identified as points outside the whiskers:

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR$$

Why Quartiles Are Important in Data Discovery (2.5 Marks)

- **Robust to Outliers:** Unlike the mean and standard deviation, quartiles are non-parametric and resistant to extreme values, providing an honest look at typical ranges.
- **Reveals Data Skewness:** Comparing $(Q_2 - Q_1)$ with $(Q_3 - Q_2)$ immediately reveals distribution asymmetry (skewness) without complex computations.
- **Standardized Outlier Detection:** *IQR*-based filtering provides a mathematically clear, automated way to flag anomalies before model training.
- **Compact Dataset Comparison:** Box plots allow rapid side-by-side comparisons of multiple feature distributions across different classes or categories.

Question 4: What qualities define data quality? Why do they matter? [5 Marks]

Qualities that Define Data Quality (2.5 Marks)

Data quality is evaluated across several fundamental dimensions:

- **Accuracy:** Whether values reflect the true, correct real-world phenomena without recording or sensor errors.
- **Completeness:** The degree to which all required data is recorded without missing fields or missing records.
- **Consistency:** The absence of contradictions across multiple tables, systems, or representations (e.g., matching address records across customer service and billing).
- **Timeliness (Currency):** How up-to-date the data is relative to the task or decision-making window.
- **Uniqueness:** The absence of duplicate records or entities in the dataset.
- **Validity (Believability / Interpretability):** Whether data adheres to defined business formats, data types, and realistic constraint ranges.

Why Data Quality Matters (2.5 Marks)

- **Garbage In, Garbage Out (GIGO):** Even the most sophisticated machine learning algorithms produce faulty, misleading predictions if trained on noisy or low-quality data.
- **Trustworthy Strategic Decisions:** Faulty metrics lead management or automated systems to make incorrect operational and financial choices.

- **Reduced Data Prep Overhead:** Data cleaning often consumes up to 80% of a data scientist's project time; high data quality dramatically cuts preprocessing and engineering costs.
 - **Regulatory Compliance & Risk Reduction:** In regulated sectors like finance and healthcare (e.g., GDPR, HIPAA), inaccurate or non-auditable data leads to severe legal penalties and reputational damage.
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User prompt: same for this

Response: Question 1: What do you understand by "frequent itemset mining"? [5 Marks]

Definition and Core Idea (2.5 Marks)

- **Frequent Itemset Mining (FIM):** A foundational data mining technique aimed at discovering sets of items (itemsets) that appear together frequently in a transactional dataset.
- **Itemset:** A collection of one or more items, denoted as $I = \{i_1, i_2, \dots, i_k\}$ (k -itemset contains k items).
- **Threshold Condition:** An itemset is considered **frequent** if its occurrence frequency (support count) satisfies a user-defined minimum threshold, known as **Minimum Support (min_sup)**:

$$\text{Support}(I) \geq \text{min_sup}$$

Key Concepts, Importance, and Extensions (2.5 Marks)

- **Role in Association Rule Mining:** FIM is the first and computationally most expensive step in discovering association rules of the form $X \Rightarrow Y$.
- **The Combinatorial Challenge:** For a dataset with d unique items, the total number of candidate itemsets is $2^d - 1$. FIM algorithms efficiently search this exponential search space without evaluating every combination.
- **Compact Representations:**
 - **Maximal Frequent Itemset:** A frequent itemset for which none of its immediate supersets are frequent.
 - **Closed Frequent Itemset:** A frequent itemset where none of its proper supersets have the same support count (preserves exact support information).
- **Common Algorithms:** Apriori, FP-Growth (Frequent Pattern Growth), and ECLAT.

Question 2: Briefly explain the concept of market basket modeling and its applications. [5 Marks]

Concept of Market Basket Modeling (2.5 Marks)

- **Concept:** Market Basket Modeling (or Market Basket Analysis) is an unsupervised data mining method that analyzes the purchasing habits of customers by finding associations between the items placed in their shopping "baskets."
- **Data Structure:** Operates on transactional databases where each record represents a transaction (e.g., receipt TID) containing a list of distinct items bought together.
- **Mechanism:** It seeks rules of the form $A \Rightarrow B$ (e.g., $\{\text{Bread, Butter}\} \Rightarrow \{\text{Milk}\}$), suggesting that a customer who buys itemset A is also likely to buy itemset B .

Applications (2.5 Marks)

- **Store Layout & Shelf Placement:** Placing frequently co-purchased items either close together (for customer convenience) or far apart (to force customers to walk past other merchandise).
- **Cross-Selling & Bundling:** Creating targeted product bundles or promotional combos (e.g., offering a discount on phone cases when buying a smartphone).
- **Recommendation Engines:** Powering "Customers who bought this also bought..." features on e-commerce platforms like Amazon.
- **Inventory Management:** Forecasting demand for complementary items simultaneously so that stock shortages of one item do not hurt sales of another.
- **Healthcare & Fraud Detection:** Identifying co-occurring symptoms/diseases in medical records or detecting typical combinations of fraudulent transaction behaviors.

Question 3: How does the Apriori algorithm handle candidate generation and pruning? [5 Marks]

The Apriori algorithm uses a level-wise iterative search where $(k - 1)$ -itemsets are used to explore k -itemsets. It relies heavily on the **Apriori Property** (monotonicity of support): *All non-empty subsets of a frequent itemset must also be frequent.*

1. Candidate Generation ($L_{k-1} \times L_{k-1} \rightarrow C_k$) (2.5 Marks)

To generate candidate k -itemsets (C_k) from the list of frequent $(k - 1)$ -itemsets (L_{k-1}), Apriori performs a **Join Step**:

- The items within each frequent itemset are maintained in lexicographical order.
- Two itemsets $l_1, l_2 \in L_{k-1}$ are joined if their first $(k - 2)$ items are identical, and they differ only in their last item:

$$(l_1[1] = l_2[1]) \wedge (l_1[2] = l_2[2]) \wedge \dots \wedge (l_1[k - 2] = l_2[k - 2]) \wedge (l_1[k - 1] < l_2[k - 1])$$

- Merging l_1 and l_2 produces a candidate k -itemset:

$$\{l_1[1], l_1[2], \dots, l_1[k - 1], l_2[k - 1]\}$$

2. Candidate Pruning (2.5 Marks)

Before scanning the database to calculate support counts, candidates in C_k are filtered using the **Prune Step**:

- **Pruning Principle:** If any $(k - 1)$ -subset of a candidate k -itemset is **not** present in L_{k-1} , that candidate cannot possibly be frequent.
- **Action:** The algorithm checks all $\binom{k}{k-1} = k$ subsets of size $(k - 1)$ for each candidate in C_k . If any subset is missing from L_{k-1} , the candidate is immediately pruned (deleted).
- **Outcome:** This dramatically shrinks the size of C_k , significantly reducing the number of costly full-database scans needed to compute the actual support of the remaining candidates.

Question 4: Define support and confidence. [5 Marks]

In association rule mining, rules are expressed in the form $A \Rightarrow B$, where A is the antecedent and B is the consequent, and $A \cap B = \emptyset$.

1. Support (2.5 Marks)

- **Definition:** Support measures how frequently an itemset or a rule appears across the entire dataset. It represents the probability that a transaction contains both A and B .
- **Formula:**

$$\text{Support}(A \Rightarrow B) = P(A \cup B) = \frac{\sigma(A \cup B)}{N}$$

Where:

- $\sigma(A \cup B)$ is the frequency/count of transactions containing both itemset A and itemset B .
- N is the total number of transactions in the dataset.
- **Significance:** Ensures statistical significance; rules with very low support may occur purely by random chance and are not practically useful.

2. Confidence (2.5 Marks)

- **Definition:** Confidence measures how often items in B appear in transactions that are known to contain A . It represents the conditional probability that a customer purchases B , given that they have already purchased A .
- **Formula:**

$$\text{Confidence}(A \Rightarrow B) = P(B \mid A) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} = \frac{\sigma(A \cup B)}{\sigma(A)}$$

- **Significance:** Measures the operational reliability and predictive strength of the rule. A high confidence value indicates a strong likelihood of the consequent occurring when the antecedent is present.

User prompt: also for this

Response: Question 1: Describe a few advantages of unsupervised learning over supervised learning. [5 Marks]

Supervised learning requires labeled target variables ($X \rightarrow y$), whereas unsupervised learning extracts underlying patterns directly from unlabeled input data (X). Key advantages include:

- **No Costly or Tedious Labeling Required (1.5 Marks):** Labeling large datasets demands substantial manual labor, human domain expertise, and high financial costs (e.g., expert medical image annotation). Unsupervised learning works directly on raw, uncured data.
- **Discovery of Unknown & Hidden Patterns (1.5 Marks):** Supervised learning is limited to predicting predefined target classes. Unsupervised techniques (clustering, association mining) can uncover unexpected groupings, hidden trends, and behavioral patterns that humans did not know existed.
- **Effective Exploratory Data Analysis & Dimensionality Reduction (1.0 Mark):** Algorithms like PCA (Principal Component Analysis) and t-SNE reduce high-dimensional spaces down to 2D/3D for visualization, data compression, and noise removal before further modeling.
- **Ideal for Real-Time Anomaly Detection (1.0 Mark):** In applications like fraud detection or network intrusion, new types of fraud constantly emerge that have never been seen or labeled before. Unsupervised density-based methods detect these purely as deviations from normal baseline behavior.

Question 2: Briefly write about information and expected information. [5 Marks]

In decision tree induction (such as the ID3 and C4.5 algorithms), both concepts originate from Claude Shannon's Information Theory to quantify impurity and measure feature importance.

1. Information / Self-Information (2.0 Marks)

- **Concept:** Quantifies the "surprise" or amount of uncertainty associated with a single event or class outcome c_i . The rarer an event, the more information it conveys when it occurs.
- **Formula:**

$$I(s_i) = -\log_2(P(c_i))$$

Where $P(c_i)$ is the probability that an arbitrary tuple belongs to class c_i . If $P(c_i) = 1$, the outcome is certain and provides zero new information ($I = 0$).

2. Expected Information (Entropy) (3.0 Marks)

- **Expected Information of a Dataset $\text{Info}(D)$ / Entropy:** Measures the average amount of information needed to identify the class label of a random tuple in a dataset D with m distinct classes:

$$\text{Info}(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

Where p_i is the non-zero probability that an arbitrary tuple in D belongs to class C_i .

- If a node is completely homogeneous (all tuples belong to one class), $\text{Info}(D) = 0$ (minimum impurity).
- If classes are equally divided, $\text{Info}(D)$ reaches its theoretical maximum (maximum impurity/uncertainty).
- **Expected Information After a Split on Attribute A ($\text{Info}_A(D)$):** The weighted average entropy required to classify a tuple across all partitions D_j produced by splitting on attribute A :

$$\text{Info}_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

- **Significance:** The difference, $\text{Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$, is the **Information Gain**, which determines the best attribute to split on at each decision tree node.

Question 3: Why do you need a hyperplane in the SVM? [5 Marks]

Role and Definition of the Hyperplane (2.5 Marks)

- **Decision Boundary:** In a Support Vector Machine (SVM), the hyperplane is the mathematical boundary that partitions the feature space into distinct decision regions corresponding to different classes (e.g., $+1$ and -1).
- In 2D space, the hyperplane is a **line** ($w_1x_1 + w_2x_2 + b = 0$).
- In 3D space, it is a **flat plane** ($w_1x_1 + w_2x_2 + w_3x_3 + b = 0$).
- In n -dimensional space, it is an $(n - 1)$ -dimensional **affine subspace**:

$$w^T x + b = 0$$

- **Classification Rule:** A new data point x^* is classified by evaluating the sign of the decision function:

$$f(x^*) = \text{sign}(w^T x^* + b)$$

Why the "Optimal" Hyperplane is Needed (2.5 Marks)

- **Maximum Margin Separation:** While infinitely many hyperplanes can separate linearly separable classes, SVM specifically searches for the **Maximum-Margin Hyperplane**—the one that maximizes the geometric distance between the hyperplane and the closest training points (**Support Vectors**).
- **Minimizing Generalization Error:** According to Structural Risk Minimization (SRM), maximizing the margin ($\frac{2}{\|w\|}$) reduces the capacity of the model, preventing overfitting and ensuring strong generalization performance on unseen test data.
- **Separation via Kernel Trick:** For non-linearly separable data, SVM maps inputs into a higher-dimensional space where a linear hyperplane can achieve clean separation without explicitly computing coordinates in that high-dimensional space.

Question 4: Do you think that multiple linear regression is better than linear regression? [5 Marks]

Core Verdict & Distinction (1.5 Marks)

Multiple Linear Regression (MLR) is **not strictly better** in every scenario; rather, it is a generalization of Simple Linear Regression (SLR):

- **Simple Linear Regression (SLR):** Models the relationship between a single independent variable X and a continuous dependent variable Y :

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- **Multiple Linear Regression (MLR):** Models the relationship between two or more independent variables ($X_1, X_2, \dots, X_p$) and a continuous dependent variable Y :

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon$$

When Multiple Linear Regression is Superior (2.0 Marks)

- **Captures Real-World Complexity:** Real-world phenomena are rarely governed by a single factor (e.g., house prices depend on size, location, age, and number of bedrooms, not just square footage).
- **Controls for Confounding Variables:** MLR isolates the specific effect of each independent variable while holding other variables constant.
- **Higher Explanatory Power (R^2):** Incorporating relevant predictors explains more variance in the response variable, reducing residual error and improving predictive accuracy.

When Simple Linear Regression is Preferable / MLR Trade-offs (1.5 Marks)

- **Risk of Overfitting:** Adding excessive or irrelevant predictors in MLR artificially inflates the training R^2 while degrading test accuracy.
- **Multicollinearity:** When independent variables are highly correlated with one another, coefficient estimates become unstable and difficult to interpret.
- **Interpretability & Data Requirements:** SLR is far simpler to visualize, compute, and explain to non-technical stakeholders, and requires significantly fewer observations to satisfy statistical assumptions.

Multiple Linear Regression is the superior choice when modeling complex, multi-factor relationships—provided multicollinearity is managed and sample size is sufficient—whereas Simple Linear Regression is preferred when a single clear predictor dominates or maximum simplicity is desired.

User prompt: finally this

Response: Question 1: How is clustering different from classification? [5 Marks]

Clustering and classification are fundamental data mining tasks belonging to different learning paradigms.

Parameter	Classification	Clustering
Learning Type	Supervised learning (learns from labeled training data).	Unsupervised learning (finds inherent structures in unlabeled data).
Target Variable	Requires predefined classes/labels (y) to map inputs ($X \rightarrow y$).	No predefined classes; groups are discovered solely based on feature similarity.
Primary Goal	Predict the class label of new, unseen data instances.	Discover natural groupings, segments, or patterns within a population.
Data Partitioning	Divides data according to decision boundaries derived from known labels.	Groups data such that intra-cluster similarity is maximized and inter-cluster similarity is minimized.
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, ROC-AUC (using ground truth).	Silhouette Coefficient, Davies-Bouldin Index, Dunn Index, Within-Cluster Sum of Squares (WCSS).
Common Algorithms	Decision Trees, Support Vector Machines (SVM), Naive Bayes, Logistic Regression.	K-Means, DBSCAN, Agglomerative Hierarchical Clustering, Gaussian Mixture Models (GMM).

Question 2: How many ways can data be clustered? [5 Marks]

Data can be clustered along several dimensions based on membership criteria, cluster relationships, and coverage:

- **1. Hard (Exclusive) vs. Soft (Fuzzy / Overlapping) Clustering (1.5 Marks):**
 - **Hard Clustering:** Each data point is strictly assigned to exactly one cluster (e.g., standard K-Means). If $x_i \in C_1$, then $x_i \notin C_2$.
 - **Soft / Fuzzy Clustering:** A data point can belong to multiple clusters simultaneously with varying degrees of membership or probabilities summing to 1.0 (e.g., Fuzzy C-Means, Gaussian Mixture Models).
- **2. Partitional vs. Hierarchical Clustering (1.5 Marks):**
 - **Partitional Clustering:** Directly divides the data into a single set of non-overlapping, flat partitions (e.g., K-Means, K-Medoids).
 - **Hierarchical Clustering:** Produces a nested tree of clusters called a **dendrogram**. It operates in two ways:
 - *Agglomerative (Bottom-up):* Starts with each point as its own cluster and progressively merges pairs.
 - *Divisive (Top-down):* Starts with all points in one mega-cluster and splits them recursively.
- **3. Complete vs. Partial Clustering (1.0 Mark):**
 - **Complete Clustering:** Forces every single object in the dataset to be assigned to some cluster.
 - **Partial Clustering:** Allows data points that represent noise, anomalies, or background outliers to remain unassigned (e.g., DBSCAN).
- **4. Monothetic vs. Polythetic Clustering (1.0 Mark):**
 - **Monothetic:** Clusters are formed by evaluating one attribute at a time (often used in simple decision-rule trees).
 - **Polythetic:** Clusters are formed by assessing all attributes simultaneously using distance or similarity metrics.

Question 3: How many clustering models are there? [5 Marks]

In data mining, clustering models are classified based on the mathematical formulation used to define a "cluster":

- **1. Centroid-Based Models (Partitioning Methods) (1.0 Mark):**
 - **Concept:** Clusters are represented by a central vector (mean or medoid). Points are assigned to the nearest center.
 - **Examples:** K-Means, K-Medoids (PAM), CLARANS.

- **Strengths/Limitations:** Efficient for large, spherical datasets; struggles with arbitrary shapes and varying densities.
- **2. Density-Based Models (1.0 Mark):**
 - **Concept:** Defines clusters as dense continuous regions of data points separated by low-density areas.
 - **Examples:** DBSCAN, OPTICS, Mean-Shift.
 - **Strengths/Limitations:** Can detect arbitrary-shaped clusters and effectively filter out noise/outliers; sensitive to parameter choices in varying densities.
- **3. Connectivity-Based (Hierarchical) Models (1.0 Mark):**
 - **Concept:** Clusters are formed based on distance connectivity between points or groups of points across a scale.
 - **Examples:** BIRCH, CURE, Agglomerative/Divisive algorithms.
 - **Strengths/Limitations:** Does not require specifying the number of clusters (k) upfront; computationally expensive ($O(n^2)$ to $O(n^3)$).
- **4. Distribution-Based (Probabilistic) Models (1.0 Mark):**
 - **Concept:** Assumes data is generated by a mixture of underlying probability distributions (commonly Gaussian distributions).
 - **Examples:** Gaussian Mixture Models (GMM) using Expectation-Maximization (EM).
 - **Strengths/Limitations:** Captures cluster covariance and provides soft membership probabilities; prone to overfitting with small sample sizes.
- **5. Grid-Based and Subspace Models (1.0 Mark):**
 - **Concept:** Quantizes the multi-dimensional space into a finite number of cells forming a grid structure; clustering operations are performed directly on the grid cells.
 - **Examples:** STING, WaveCluster, CLIQUE.
 - **Strengths/Limitations:** Highly scalable runtime independent of the number of data objects; cell boundaries can fragment natural clusters.

Question 4: What are the challenges in high dimension data clustering? [5 Marks]

Clustering high-dimensional data (e.g., gene expressions, text documents, image descriptors) poses severe mathematical and computational challenges, collectively known as the **Curse of Dimensionality**:

- **Distance Metric Degeneration / Concentration of Distances (1.5 Marks):**
 - As the dimensionality $d \rightarrow \infty$, the contrast between the distance to the nearest neighbor and the distance to the furthest neighbor vanishes:

$$\lim_{d \rightarrow \infty} \frac{\text{dist}_{\max} - \text{dist}_{\min}}{\text{dist}_{\min}} \rightarrow 0$$

- All pairs of points become almost equidistant from one another, rendering Euclidean and Manhattan distance metrics meaningless.
 - **Data Sparsity (Empty Space Phenomenon) (1.0 Mark):**
 - The volume of the feature space increases exponentially with each added dimension ($V \propto r^d$). The data points become extremely sparse, and the concept of a high-density cluster collapses because points rarely reside near each other.
 - **Irrelevant and Noisy Attributes (1.0 Mark):**
 - In datasets with hundreds or thousands of features, clusters often exist only within small, localized subsets of dimensions (**subspaces**). Irrelevant features act as random noise that masks meaningful cluster structures.
 - **Exponential Computational & Memory Complexity (1.0 Mark):**
 - Calculating pairwise distance matrices and evaluating multi-dimensional density functions scales poorly with dimension count, causing standard algorithms like DBSCAN or Hierarchical clustering to run out of memory or time.
 - **Mitigation Strategies (0.5 Marks):**
 - To cluster high-dimensional data effectively, practitioners must use **Dimensionality Reduction** (PCA, Autoencoders), **Feature Selection**, or specialized **Subspace/Projected Clustering** algorithms (e.g., CLIQUE, PROCLUS).
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